

Telemetry Schema Data Dictionary

1. Introduction & Core Architecture

This technical data dictionary acts as the master schema definition manual for the Compliance Posture Assessment Suite. It comprehensively details all 18 evaluated technical data points, outlining every collected telemetry field, data types, security relevance, and their integration with the underlying automated Severity Classification and Regulatory Compliance Mapping Matrices (PCI DSS v4.0, NIST SP 800-53 Rev. 5, and the DPDP Act 2023).

2. Complete Telemetry Schema Reference Catalog

Datapoint 01: 01_installed_software

FRAMEWORKS: PCI-6.2, NIST CM-8, DPDP-SEC-01 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_SOFTWARE

Inventories all software products currently registered within the 32-bit and 64-bit local system uninstallation registry subkeys to evaluate software integrity and detect unauthorized tools.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
DisplayName	The full readable name of the installed software program. Cross-referenced against the application blacklist.
DisplayVersion	The specific release build version string used to trace security patching levels.
Publisher	The organization or vendor providing the software certificate wrapper.
InstallDate	The recorded system timestamp when the application was written to the host.
InstallLocation	The direct path inside storage hives hosting the binaries.
EstimatedSizeMB	Calculated application footprint footprint on the local storage partition.
DigitalSignature	The cryptographic status verification of the app installer file.
RiskLevel	Categorized scale (LOW, MEDIUM, HIGH) mapping risk indices dynamically based on threat profiles.

Datapoint 02: 04_hardware_inventory

FRAMEWORKS: PCI-9.1, NIST CM-8, DPDP-SEC-01 | DEFAULT INGESTION TIER: INFO

Harvests baseline architecture parameters directly from the native CIM/WMI schema classes to form immutable device identification blocks.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Manufacturer	The corporate entity producing the hardware chassis (e.g., Dell Inc., HP).
Model	The commercial model layout used to track asset inventory baselines.
Name	The current network hostname configured for the node.
NumberOfProcessors	The hardware count of physical processor sockets inside the core ring.
NumberOfLogicalProcessors	Total multi-threaded core slots visible to the operating system kernel.
TotalPhysicalMemory	Total installed hardware RAM byte count used to establish asset parameters.
SystemType	Architectural classification mapping (e.g., x64-based PC).
SystemFamily	The manufacturing classification tier assigned by the system integrator.
Domain	The active Active Directory domain membership or local workgroup boundary.
PrimaryOwnerName	The structural administrative registration assignment marker.
BootupState	Current startup configuration mode (e.g., Normal boot, Safe mode).

Datapoint 03: 05_windows_services

FRAMEWORKS: PCI-2.2, NIST CM-7, DPDP-MON-01 | DEFAULT INGESTION TIER: INFO

Tracks daemon execution integrity across core background system tasks to ensure mandatory protection wrappers remain functional.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Name	The system registry identifier used to invoke daemon executions (e.g., WinDefend).
DisplayName	The readable description shown inside monitoring management interfaces.
State	Active execution condition variables (e.g., Running, Stopped).
StartMode	The boot invocation parameter profile (e.g., Automatic, Manual, Disabled).
StartName	The security identity profile hosting the service environment (e.g., LocalSystem).
PathName	The exact file argument line executed by the loader to mount the tool.

Datapoint 04: 06_failed_logins

FRAMEWORKS: PCI-10.2, NIST AC-7, DPDP-AUD-01 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_FAILED_LOGINS

Parses Windows Security log records matching event schema maps to identify anomalies, credential stuffing, and active brute-force attempts.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
TimeCreated	The specific UTC timestamp detailing when the authentication failure occurred.
Id	The static event log verification identifier (always 4625 for failed logins).
TargetUserName	The user account profile targeted by the failed login request.
SourceIPAddress	The network origin IP address generating the unauthenticated traffic block.
WorkstationName	The source device hostname passing access negotiation metadata.
LogonType	The mechanism index defining the entry method (e.g., Type 2 Interactive, Type 3 Network).
FailureReason	The sub-status error code detailing why negotiation failed (e.g., bad password).
StatusCode	The NTSTATUS error sequence extracted directly from the event kernel envelope.

Datapoint 05: 07_successful_logins

FRAMEWORKS: PCI-8.1, NIST AC-2, DPDP-ACC-01 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_SUCCESSFUL_LOGINS

Monitors authorized access patterns across all authentication interfaces to trace privilege escalation and unusual lateral movements.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
TimeCreated	The timestamp showing when access was successfully granted to the profile.
TargetUserName	The exact user identifier registering the active session instance.
SourceIPAddress	The network address location that generated the login signal.
Workstation	The caller system metadata signature transmitted during negotiation loops.
LogonType	The access method context code tracking session configurations (e.g., 10 RDP).
AuthenticationPackage	The subsystem handler module performing validation (e.g., NTLM, Kerberos).
ElevatedToken	Boolean variable flagging if the session carries elevated admin tokens.

Datapoint 06: 10_firewall_configuration

FRAMEWORKS: PCI-1.2, NIST SC-7, DPDP-SEC-02 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_FIREWALL

Examines network filtering profiles to track the operational state of local software firewall systems.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
FirewallEnabled	Boolean indicator defining if network firewall mechanisms are active.
ComplianceStatus	The engine-calculated status flag (PASS/FAIL) requiring all profiles to be active.
InboundRuleCount	Total number of filtering conditions regulating incoming traffic vectors.
OutboundRuleCount	Total number of rules tracking connections initialized by local binaries.
InboundRules / OutboundRules	Sub-array list properties containing rule specific configurations: RuleName, Direction, Action, Protocol, LocalPort, RemotePort, and Enabled.

Datapoint 07: 12_registry_autoruns

FRAMEWORKS: PCI-11.5, NIST SI-7, DPDP-MON-02 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_AUTORUNS

Scans system startup registry keys to detect persistent execution methods planted by malicious actors.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Path	The location of the evaluated registry key (e.g., HKLM:\...\Run).
Executable	The program path or command executed when a user authenticates.
Publisher	The software author attribute extracted from the target file signature block.
SignatureStatus	The cryptographic state validation (e.g., Signed, Unsigned, Invalid).
Hash	The SHA-256 signature code of the file pointed to by the auto-start entry.

Datapoint 08: 13_scheduled_tasks

FRAMEWORKS: PCI-11.5, NIST SI-7, DPDP-MON-02 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_TASKS

Audits active system tasks to identify hidden schedules or suspicious binaries configured for automated background execution.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
TaskName	The name assigned to the automated background task configuration.
TaskPath	The internal folder location where the task configuration resides.
State	Current operational lifecycle mode (e.g., Ready, Running, Disabled).
Author	The metadata account attribute showing which user identity created the task.
LastRunTime	Timestamp tracking when the script or binary was last invoked.
NextRunTime	Timestamp tracking the next planned execution loop.
Action	The command or application line triggered when the task condition is met.
RunAsUser	The security identity context used to execute the underlying task binary.
Hidden	Boolean property flagging if the configuration is omitted from standard UI tools.

Datapoint 09: 16_user_accounts_and_privileges

FRAMEWORKS: PCI-7.1, NIST AC-6, DPDP-ACC-02 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_ACCOUNTS

Tracks local user profile accounts and group privileges to prevent dormant account exploitation and administrative group dilution.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
UserName / Name	The unique local profile name registered on the host platform.
AccountType	The authorization level category assigned to the account (e.g., Admin, Standard).
Enabled	Boolean variable indicating if the profile account is active and available for use.
LastLoginDate / LastLogon	The timestamp tracking when the user profile last opened a session.
PasswordLastSet	Timestamp recording when the user credentials were last modified.
DaysInactive	Calculated metric tracing total elapsed downtime days since the last logon event.
AccountStatus	Calculated baseline categorization (ACTIVE / INACTIVE) evaluating profile currency.

Datapoint 10: 17_windows_defender_status

FRAMEWORKS: PCI-5.1, NIST SI-3, DPDP-DPC-01 | DEFAULT INGESTION TIER: INVESTIGATE

Queries antimalware engine states to verify that core host-level real-time threat protection layers are fully active.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
AMProductVersion	The active software platform version engine identifier.
DefenderSignaturesOutOfDate	Boolean variable flagging if security protection definitions are expired.
RealTimeProtectionEnabled	Boolean verification flag ensuring ongoing behavioral file-scanning is active.
AntivirusEnabled	Core parameter tracking if anti-malware tracking subsystems are on.
AntispywareEnabled	Subkey boolean tracking the status of local peripheral monitoring agents.
IsTamperProtected	Enforces signature preservation and locks features against administrative degradation.
BehaviorMonitorEnabled	Tracks live script behaviors to block execution signatures in memory.

Datapoint 11: 20_boot_shutdown_events

FRAMEWORKS: PCI-10.6, NIST AU-5, DPDP-AUD-02 | DEFAULT INGESTION TIER: STATISTICAL

Extracts system log lifecycle records to document host operational availability profiles and track unexpected reboots.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
TimeCreated	The specific event timestamp written during hardware state changes.
Id	The operating system identifier key (6005 marks Startup, 6006 marks Shutdown).
ProviderName	The source kernel infrastructure tracking component (Microsoft-Windows-Eventlog).
MachineName	The local host device name writing the log record to the local drive.

Datapoint 12: 21_audit_policy_configuration

FRAMEWORKS: PCI-10.1, NIST AU-6, DPDP-AUD-03 | DEFAULT INGESTION TIER: INFO

Audits detailed system policy configurations to confirm the continuous generation of log files across security boundaries.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Configuration	Nested data array containing sub-properties: Category, Subcategory, and Setting (e.g., Success and Failure tracking logs).
RecentAuditEvents	A structure recording the 20 most recent logs written to the Security event channel, capturing properties like LogName, EventID, Level, and User.

Datapoint 13: 22_drivers_inventory

FRAMEWORKS: PCI-2.2, NIST SI-4, DPDP-MON-03 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_DRIVERS

Maps active kernel-level extensions and peripheral device drivers to uncover unauthorized hardware control software.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
DeviceName	The system-assigned hardware descriptor name for the driver instance.
DriverVersion	The publisher version string used to identify vulnerable driver components.
IsSigned	Boolean property verifying if the module has a valid signature. Unsigned kernel drivers present critical security risks.
DriverHash	The cryptographic signature code computed for binary execution verification.
Signer	The corporate entity backing the driver certificate wrapper (e.g., Microsoft, Intel).
DriverAgeDays	Elapsed days since the driver file creation date to flag old legacy objects.
KernelMode	Boolean variable indicating if the driver operates with ring 0 core privileges.

Datapoint 14: 23_more_windows_settings

FRAMEWORKS: PCI-2.2, NIST CM-6, DPDP-SEC-03 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_WINDOWS_SETTINGS

Aggregates core configuration metrics to calculate an overall hardening score across subkey boundaries.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Control	Static control index code assignment mapping target indicators (WIN-SET).
Status	The evaluated compliance summary label (PASS / FAIL) based on hardening scores.
Score	Calculated score value (0-100) reflecting system posture across properties like UAC, LSA, and RDP settings.

Datapoint 15: 24_usb_direct_connection

FRAMEWORKS: PCI-3.4, NIST MP-7, DPDP-DPC-02 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_USB

Audits active physical interface connections to detect insider threats and unauthorized data exfiltration attempts.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Name	The descriptive hardware string of the attached peripheral device.
Manufacturer	The corporate entity or vendor manufacturing the peripheral device.
DeviceID	The hardware interface hardware identifier string written to the OS bus.
VID	The 4-digit Vendor ID code used to identify approved hardware vendor classes.
PID	The 4-digit Product ID code tracking device manufacturing types.
Vendor	The manufacturer identity string verified during connection sweeps.
FirstSeen / LastSeen	Timestamps documenting connection durations on the host interface.
DriverName	The system service engine driving the hardware bus (e.g., USBSTOR).
Class	Device class categorization signatures (e.g., USB, Media, HID).
IsStorageDevice	Boolean flag identifying if the peripheral is an external storage block.
IsAuthorized	Boolean checking against the approved VID list to flag unauthorized devices.

Datapoint 16: 25_bios_snapshot

FRAMEWORKS: PCI-2.2, NIST CM-8, DPDP-SEC-04 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_BIOS_SNAPSHOT

Evaluates low-level firmware and hardware state metrics to establish foundational platform security levels.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
Manufacturer	The firm issuing the system firmware image (e.g., Dell, Insyde).
SMBIOSBIOSVersion	The release version sequence used to audit motherboard patch levels.
SecureBootEnabled	Boolean verification ensuring the system boots using only signed, verified kernel parameters.
TPM_Present	Boolean verifying the installation of a physical Trusted Platform Module.
TPM_Version	The revision level of the local cryptographic coprocessor (e.g., 2.0).
BIOS_Age_Days	Total days since firmware release to track legacy compliance exposure.
Firmware_Compliance	Master validation status flag ensuring complete firmware security readiness.

Datapoint 17: 26_windows_scan_history

FRAMEWORKS: PCI-5.2, NIST SI-3, DPDP-MON-04 | ENGINE PROCESSING: SEVERITYENGINE.PROCESS_SCAN_HISTORY

Tracks endpoint security scan histories and threat logs to identify unresolved infections on the host system.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
QuickScanStart / QuickScanEnd	Timestamps logging operational run times for incremental defender evaluations.
LastThreatDetected	Timestamp documenting the host's most recent active malware detection event.
ThreatCount	Total number of malicious objects caught by local scanning filters.
ThreatNames	Array string listing the signatures of identified malicious files.
RemediationStatus	Tracks cleaning actions to flag unresolved or active infection vectors (e.g., Success, Failed).
EngineVersion	The release version of the local scanner core.
SignatureVersion	The release version tracking database signature update stamps.

Datapoint 18: 27_usb_setting_history

FRAMEWORKS: PCI-3.4, NIST MP-7, DPDP-DPC-03 | DEFAULT INGESTION TIER: INVESTIGATE

Audits system registry logs to discover past peripheral connections, preventing historical data loss incidents.

FIELD NAME	FIELD EXPLANATION & SECURITY RELEVANCE
FriendlyName	The descriptive hardware label saved by the system upon the device's first attachment.
DeviceDesc	The internal system description string mapping the peripheral's driver class.
Mfg	The manufacturing entity name extracted from the device's descriptors.
HardwareID	The comprehensive bus architecture string used for precise driver assignment.